WORKED SOLUTION

STEP II 2017, Question 11

Mechanics • Projectile

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Read the question before you go on. It is in the STEP Support Programme's question database:

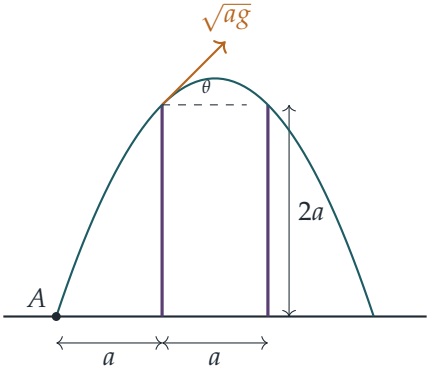
step.maths.org/questions/17-s2-q11

The key idea

Once the particle is level with the top of the first wall, the rest of its flight is a fresh projectile problem: it starts at height $2a$ with a known speed and must still be at least at height $2a$ a horizontal distance a later. Energy gives that speed without knowing the angle of projection. Writing the trajectory in terms of $\tan\theta$ turns "can it clear the second wall?" into "does a quadratic in $\tan\theta$ have a real root?", which handles both parts at once.

Solution

Model each particle as a point mass moving freely under gravity, with no air resistance. Take x horizontal (towards the walls) and y vertically up.



Part (i)

Speed over the first wall. Only gravity does work, so kinetic plus potential energy is constant. Rising from the ground to height $2a$, with speed v at the top of the wall,

$$\tfrac{1}{2}m(5ag) = \tfrac{1}{2}mv^2 + mg(2a) \quad \Longrightarrow \quad v^2 = 5ag - 4ag = ag,$$

so the particle passes over the first wall with speed $\sqrt{ag}$.

The angle. Let the velocity make angle θ above the horizontal as the particle passes over the first wall; $\theta > 0$, because the particle is at height $2a$ again later (over the second wall), so it must still be rising. Measure from the top of the first wall: after horizontal distance X the height above wall-top level is

$$Y = X\tan\theta - \frac{gX^2}{2v^2\cos^2\theta} = X\tan\theta - \frac{X^2}{2a}\,(1+\tan^2\theta),$$

using $v^2 = ag$ and $\sec^2\theta = 1 + \tan^2\theta$. Just clearing the second wall means $Y = 0$ at $X = a$:

$$a\tan\theta - \tfrac{a}{2}(1+\tan^2\theta) = 0 \iff \tan^2\theta - 2\tan\theta + 1 = 0 \iff (\tan\theta - 1)^2 = 0.$$

So $\tan\theta = 1$ and $\theta = 45^\circ$, as required. (Equivalently: the range on level ground from height $2a$ back to $2a$ is $v^2\sin 2\theta/g = a\sin 2\theta$, which equals a only when $\sin 2\theta = 1$.)

Distance of A. The horizontal velocity component never changes, and over the first wall it is $\sqrt{ag}\cos 45^\circ = \sqrt{ag/2}$; the vertical component there is also $\sqrt{ag/2}$, upwards. At A the vertical component u satisfies $u^2 = \frac{1}{2}ag + 2g(2a) = \frac{9}{2}ag$, so $u = 3\sqrt{ag/2}$. (As a check, $\frac{1}{2}ag + \frac{9}{2}ag = 5ag$, matching the launch speed.) The time from A to the first wall is

$$t = \frac{u - \sqrt{ag/2}}{g} = \frac{2\sqrt{ag/2}}{g},$$

and in that time the horizontal distance covered is

$$\sqrt{\tfrac{ag}{2}}\cdot\frac{2\sqrt{ag/2}}{g} = \frac{2\cdot\frac{1}{2}ag}{g} = a.$$

So A is a distance a from the foot of the first wall.

Part (ii)

Now the particle passes over the first wall at height $2a + h$. By energy, its speed w there satisfies

$$w^2 = 5ag - 2g(2a+h) = g(a-2h).$$

The particle moves towards the walls, so its horizontal velocity is non-zero and $w^2 > 0$; hence $h < \frac{1}{2}a$.

Let its direction there make angle ϕ with the horizontal, and write $T = \tan\phi$. Measuring from the particle's position over the first wall, after horizontal distance X its height above that point is

$$Y = XT - \frac{gX^2}{2w^2}(1+T^2) = XT - kX^2(1+T^2)/a^2, \qquad \text{where } k = \frac{ga^2}{2w^2} = \frac{a^2}{2(a-2h)}.$$

At the second wall, $X = a$, so $Y = aT - k(1+T^2)$. To clear that wall the particle must be at height at least $2a$, i.e. $Y \geq -h$:

$$aT - k(1+T^2) + h \geq 0 \iff kT^2 - aT + (k-h) \leq 0.$$

Ask yourself We don't know ϕ. How can we show that *no* direction works?

Since $k > 0$, the quadratic $kT^2 - aT + (k-h)$ opens upwards. It takes a value ≤ 0 for some real T only if its discriminant is non-negative. That discriminant is

$$\Delta = a^2 - 4k(k-h) = a^2 - \frac{a^4}{(a-2h)^2} + \frac{2a^2h}{a-2h}.$$

Multiplying by the positive quantity $(a-2h)^2/a^2$,

$$\frac{(a-2h)^2}{a^2}\Delta = (a-2h)^2 - a^2 + 2h(a-2h) = -2ah.$$

Since $a > 0$ and $h > 0$, this is negative, so $\Delta < 0$. Then $kT^2 - aT + (k-h) > 0$ for every real T, so whatever direction the particle has over the first wall, it is below height $2a$ when it reaches the second wall. It does not clear the second wall.

Note that $h = 0$ gives $\Delta = 0$ and the double root $T = a/(2k) = 1$: exactly the $45°$ of part (i).

Checks

- In part (i), the components at A are $\sqrt{ag/2}$ and $3\sqrt{ag/2}$, whose squares add to $5ag$. Stepping the motion forward in time from A with $a = 1.7\,\text{m}$ and $g = 9.81\,\text{m}\,\text{s}^{-2}$ gives heights $2.000a$ at both walls.
- The trajectory is symmetric about its highest point, which in part (i) lies halfway between the walls, at horizontal distance $1.5a$ from A. So the particle lands $3a$ from A, a distance a beyond the second wall: the picture is symmetric.
- For part (ii), a computer check over thousands of random launch angles and launch points with speed $\sqrt{5ag}$ found none that passed above the first wall and also cleared the second.

Common slips

- Forgetting that θ in part (i) must be the angle over the first wall, not the angle of projection at A (that one has $\tan\alpha = 3$).
- In part (ii), checking only one particular angle (such as $45°$). The claim is about every possible direction, which is why the discriminant is needed.

Beyond STEP

Part (ii) says that, from a point over the first wall, a particle with this energy can only just reach the top of the second wall: that wall-top lies on the boundary of what is reachable. All points reachable from a given point with a given speed lie inside a parabola called the *envelope* or *safety parabola*, and points on it are reached by exactly one trajectory, which is the double root seen here. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats projectiles and energy methods at first-year level.

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